Medhansh Garg

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Education

University of Illinois, Urbana-Champaign Aug. 2023 – May 2027 (Expected)
- B.S. in Computer Engineering | Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Distributed Systems, Computer Networks, Databases

Skills

- **Infra & Ops**: Linux, systemd, Docker, Kubernetes, AWS, Azure, nginx, Tailscale, MQTT, Samba, GitHub Actions CI/CD, Vercel
- **Languages**: Python, Go, C, C++, TypeScript/JavaScript, Java, Kotlin, Bash, SQL
- **Backend & Data**: FastAPI, Node.js, Next.js, PostgreSQL, Supabase, SQLite, Prisma, REST APIs
- **Testing & Monitoring**: pytest, Vitest, Playwright, Home Assistant, MQTT discovery, Git

Experience

Hopcharge, Gurugram, India | *Software Engineering Intern* Jun. 2026 – Aug. 2026
- Consolidated **5 manual scripts** into Grid Ops, a FastAPI/React service forecasting battery-swap demand across **3 Delhi-NCR metros**; deployed and operated behind nginx on an ARM cloud VM
- Automated cross-platform releases of VOLT.CV, a hiring dashboard, via GitHub Actions for **4 OS/architectures**; enforced database-layer access control with row-level security on Google identities
- Architected the Ad Engine, auto-generating and publishing ads to Meta and YouTube; scaled output **8x** (2–3/week to **3/day**)

CS 124 Honors, Urbana, IL | *Lead Web Developer* Aug. 2025 – Present
- Lead an **11-person** team running the course platform for **300+ students** and **30+ staff**; set up the CI/CD pipeline with a **246-test** suite (unit, integration, E2E) gating every production promotion
- Improved Lighthouse score from **76 to 99** by eliminating bloated assets and render-blocking paths; implemented CILogon SSO with 6 least-privilege roles; kept all database credentials server-side

IoT++, Remote | *Software Engineering Intern* Jun. 2024 – Aug. 2024
- Deployed real-time fire/smoke detection to **12 edge devices** in a factory pilot (Docker, Kubernetes, Azure); improved NPU inference speed **13x** and accuracy **62%** over base YOLOv8
- Trained models routing trucks to weighbridges from live traffic and equipment throughput

Care Health Insurance, Gurugram, India | *Cybersecurity Intern* Jun. 2025 – Aug. 2025
- Discovered **60+ unauthenticated internet-exposed services**; **30** moved offline after my report
- Bypassed SSL pinning and root detection in the customer Android app, enabling live traffic testing

Projects

Mainframe: Pi Homelab | *Linux, Docker, Tailscale* hackoverflow404.github.io/Portfolio/experience?entry=mainframe
- Operate **19 services** on a 2GB Raspberry Pi since May 2025 (2TB NAS, Home Assistant, MQTT broker, voice assistant, camera recorder), mostly self-written systemd units with OOM priorities protecting critical services
- Secured remote access via Tailscale mesh with **zero open router ports**; wrote monitoring services publishing Pi and laptop health to Home Assistant, flagging crashes as unavailable rather than stale
- Built SECUR-T, fanning one camera stream to Home Assistant and a recorder with **18h** of footage on the NAS at ∼**7%** Pi CPU

Pensieve: AI Note Librarian | *Python, systemd, Git* hackoverflow404.github.io/Portfolio/experience?entry=pensieve
- Deployed a Pi-hosted notes service reachable from every device over Tailscale: git snapshots every **15 min**, offsite backup, health checks alerting on service, backup or memory failures
- Automated note filing with a sandboxed AI agent (**125 notes**, **68 runs** in 2 weeks), guarded by a rule-based checker (**76 tests**) rejecting any run that alters original text

Clippy: Private Voice Assistant | *Python, Go, Kotlin* github.com/HackOverflow404/Pi-Voice-Assistant
- Developed a voice assistant with on-device speech models and a Go WhatsApp bridge; implemented a SQLite outbox for zero lost or duplicated requests across reconnects and crashes (**171 tests**)
- Diagnosed voice dropouts every **20–40s** as Wi-Fi driver power save; eliminated them with a USB link and Wi-Fi failover; patched Android's Bluetooth stack, stabilizing speaker latency at **30–180ms**

TPS-L2: Spotify Wall Display | *JavaScript, Vercel, Redis, Raspberry Pi* github.com/HackOverflow404/TPS-L2
- Built a now-playing display on a salvaged laptop screen with a serverless backend and QR sign-in for keyboardless screens; cut Redis usage ∼**10x** with signed, self-expiring codes, staying on the free tier
- Replaced Chromium and X11 with a minimal WebKit kiosk as a systemd service, freeing memory for the NAS and voice assistant

ECE 391 RISC-V Kernel | *C, RISC-V, WebAssembly* ece-391-kernel-demo.vercel.app
- Co-developed an OS kernel from scratch (VirtIO drivers, filesystem, virtual memory, scheduler, syscalls, shell with pipes); compiled an emulator to WebAssembly so it boots live in any browser
- Implemented VirtIO drivers directly from the hardware spec; debugged crashes from raw register state and interrupt races

Leadership & Involvement

Illinois MicroTech, Urbana, IL | *External Vice President* | *EOH Project Lead* | *HackerFab Automation Member* Oct. 2025 – Present
- Run external communications, overseeing the outreach team, marketing chairs, website and EOH chairs
- Led a **15-person** team building a 4-stage interactive MEMS exhibit for **20,000+** Engineering Open House visitors
- HackerFab: automating the wafer stage of a student-built chip-fabrication rig for autonomous full-wafer movement